Lab experiment-8

Experiment 8: Depth First Search (DFS)

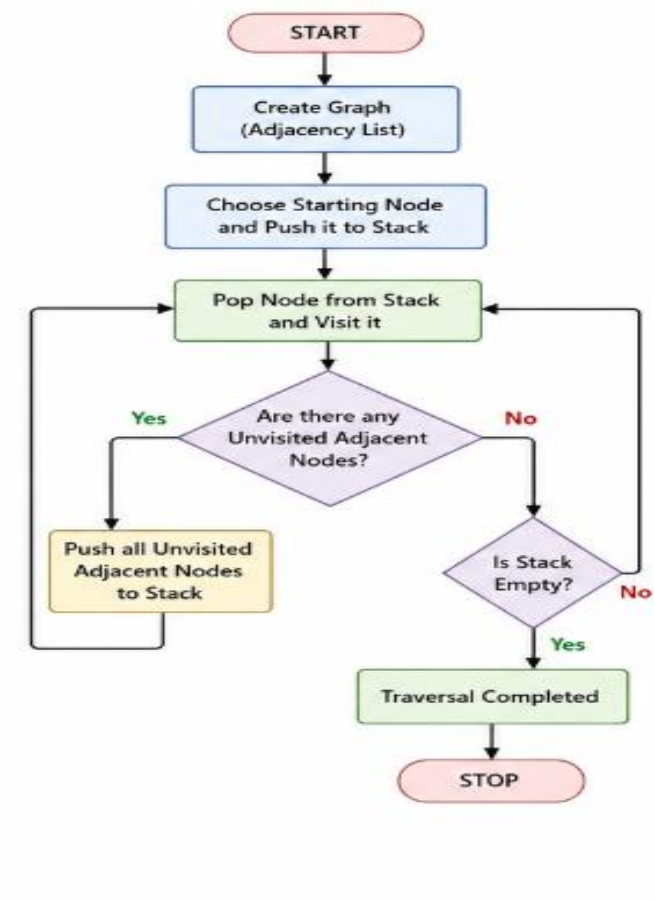
Aim

To implement the Depth First Search (DFS) algorithm using Python.

Objective

To traverse all the nodes of a graph using the DFS algorithm.

Flowchart



Algorithm

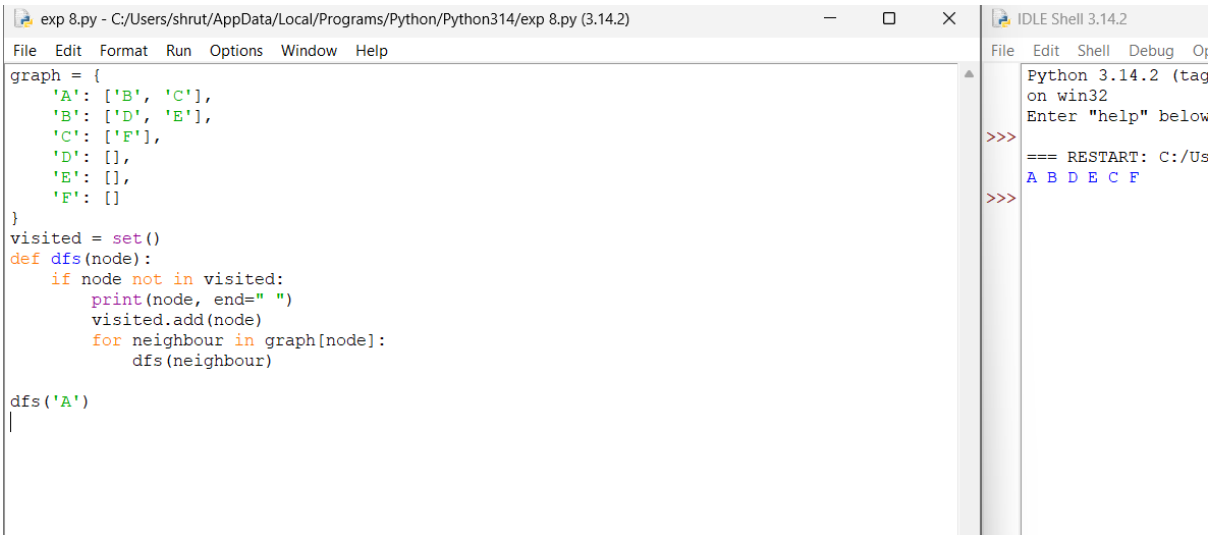
1. Create a graph using an adjacency list.
2. Mark the starting node as visited.
3. Visit the current node.
4. Recursively visit all unvisited adjacent nodes.
5. Continue until all nodes are visited.

Code

```
graph = {
   'A': ['B', 'C'],
   'B': ['D', 'E'],
   'C': ['F'],
   'D': [],
   'E': [],
   'F': []
}
visited = set()
def dfs(node):
   if node not in visited:
      print(node, end=" ")
      visited.add(node)
      for neighbour in graph[node]:
         dfs(neighbour)

dfs('A')
```

Output



Result

The graph was successfully traversed using the DFS algorithm.

Conclusion

The DFS algorithm visits nodes by exploring each branch completely before backtracking.